S 2039

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SSLC EXAMINATION, MARCH - 2025

CHEMISTRY

(English)

Time : 1½ Hours

Total Score : 40

Instructions :

- The first **15** minutes is cool-off time.
- You may use this time to read the questions and plan your answers.
- Answer only on the basis of instructions and questions given.
- Consider score and time while answering.

Score

SECTION - A

Answer any 4 questions from 1 to 5. Each question carries 1 score. 4x1=4

1. Which of the given compounds is a mineral of Aluminium ? 1
(Calamine, Cuprite, Cryolite, Magnetite)

2. Which polymer is formed from isoprene ? 1

3. Find the total number of atoms in 2 GMM water (H_2O). 1

4. Which is the product obtained when SO_3 gas is dissolved in concentrated sulphuric acid ? 1

5. Many of the metals in _____ block are used as catalysts in petroleum industry. 1
(s, p, d, f)

SECTION - B

Answer any 4 questions from 6 to 10. Each question carries 2 scores. 4x2=8

6. Classify the following compounds into alkanes, alkenes and alkynes. 2
C_2H_4, C_5H_{12}, C_3H_8, C_7H_{12}

7. Two gases Oxygen and Nitrogen at STP are given.
   (a) Calculate the number of molecules in 64 g of Oxygen. 1
   (b) Calculate the mass of Nitrogen having the same volume as that of 64 g of Oxygen. 1
   (Hint : Atomic mass of Oxygen = 16, Nitrogen = 14)

P.T.O.

Score

8. Two reactions related to extraction of metals are given. Among these reactions identify calcination and roasting.
   (a) Cu_2S ore is converted to Cu_2O by heating. 1
   (b) Carbonates and hydroxides of metals decompose to form their oxides. 1

9. Identify A and B.
   (a) $CH_3-OH+CO \xrightarrow{\text{catalyst}} \underline{\quad(A)\quad}$ 1
   (b) $CHCl_3+Cl_2 \longrightarrow \underline{\quad(B)\quad}+HCl$ 1

10. The chemical formula of a compound is C_2H_6O.
   (a) Write the structural formula of any one of the functional isomers. 1
   (b) Write the IUPAC name of this functional isomer. 1

SECTION - C

Answer any 4 questions from 11 to 15. Each question carries 3 scores. 4x3=12

11. Copper is electroplated on an iron bangle.
   (a) Which electrolyte is used here ? 1
   (b) Write any one of the advantages of electroplating. 1
   (c) Does the intensity of colour of electrolyte change during this process ? Why ? 1

12. Subshell electronic configuration of an element 'A' is given in two different ways. (Symbol is not real)
   (i) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^2$
   (ii) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2$
   (a) Which of these is the correct subshell electronic configuration ? Why ? 2
   (b) Identify the block in the periodic table to which this element belongs ? 1

13. Two reversible reactions are given.
   (I) $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)} + \text{Heat}$
   (II) $2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)} + \text{Heat}$
   (a) Identify the reaction in which the pressure has no effect. 1
   (b) How do the following factors influence the amount of product in the reaction (II) ? 2
      (i) Increase the temperature
      (ii) Increase the pressure

Score

14. Chemical equations of some reactions are given.

(i) $CH_3-CH=CH-CH_3+HBr \longrightarrow CH_3-CH_2-CHBr-CH_3$

(ii) $n\,CH_2=\underset{\underset{Cl}{|}}{CH} \longrightarrow \left[CH_2-\underset{\underset{Cl}{|}}{CH}\right]_n$

(a) Which types of reactions are these ? 1

(b) Identify the product in the reaction (ii) and write any one use of it. 2

15. Volume and number of molecules of some gases at 27°C and 2 atm pressure are given in the table.

Gas	Volume (L)	Number of molecules
Nitrogen	10	x
Carbon dioxide	(i)	2x
Oxygen	5	(ii)

(a) Complete the table. 1

(b) What will be the volume of carbon dioxide gas if the pressure is increased to 4 atm ? 2

(Temperature and number of molecules remain constant)

SECTION - D

Answer any 4 questions from 16 to 20. Each question carries 4 scores. **4x4=16**

16. Manganese (Mn) is an element that belongs to d block in the periodic table.

(a) Outer subshell electronic configuration of Mn is $3d^5 4s^2$. Find the atomic number of Manganese. 1

(b) Find the oxidation state of Mn in Mn_2O_7. 1

(c) Write period number and group number of Manganese. 1

(d) Write the subshell electronic configuration of Mn^{4+} ion. 1

(Hint : Oxidation state of Oxygen = −2)

17. A few drops of the solution 'X' is added to Magnesium sulphate solution taken in a test tube. A white precipitate is formed.

(a) What is 'X' ? 1

(b) What is the chemical name of white precipitate formed here ? 1

(c) What happens to the white precipitate when dilute hydrochloric acid is added ? 1

(d) Which type of salt is identified by this experiment ? 1

Score

18. Three metals Ag, Mg, Cu and their salt solutions are given.
    (a) How many galvanic cells can be constructed using these metals ? 1
    (b) If we construct a galvanic cell using the most reactive metal and the least reactive metal from the given metals, identify the cathode and anode. 1
    (c) Write the chemical equation of the reaction which takes place at anode and cathode in the above cell. 2

19. Iron is a very important metal in the industrial and construction field.
    (a) Write the name and chemical formula of ore of iron used in the industrial production of iron. 1
    (b) Write the chemical equation of slag formation in the industrial production of iron. 1
    (c) Which mineral of iron is known as fool's gold ? Iron pyrite 1
    (d) Write the names of two alloy steels which contain the same constituent elements. 1

20. Match the following suitably. 4

A	B	C
i) $CH_3-\underset{\displaystyle CH_3}{\underset{\displaystyle |}{CH}}-CH-CH_3$	C_3H_8O	Pentanoic acid
ii) $CH_3-CH=CH-CH_3$	i) C_6H_{14}	iii) Propan-2-ol
iii) $CH_3-\underset{\displaystyle OH}{\underset{\displaystyle |}{CH}}-CH_3$	iv) $C_5H_{10}O_2$	ii) But-2-ene
iv) $CH_3-CH_2-CH_2-CH_2-COOH$	ii) C_4H_8	i) 2,3-Dimethylbutane

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